Pratik **Joshi**

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Experience

Current – Jun. 2023 | **Google DeepMind**, Mountain View, CA
Senior Research Engineer, Post-Training for Code

- Core contributor to Gemini 4.0 (Argon), 3.5, 3.0, 2.5, 1.5 Pro, 1.0, improving code reasoning and agentic tool use in post-training.
- **Tool-use generalization:** Improved Gemini's tool use via RL and targeted SFT data: fewer tool-call hallucinations, better generalization to complex external harnesses (Antigravity, Cursor), and stronger instruction following via reward shaping.
- **Agentic tool-calling framework:** Co-created Gemini's unified framework for multi-step code RL, SFT distillation, and agentic evals; used to post-train every Gemini model since Gemini 3 and adopted by 20+ teams across Google, including Cloud and Chrome. Built sandboxed execution, failure handling, and dynamic toolbox orchestration.
- **RL captain for a Flash-size training run:** Owned a versioned run end to end, coordinating across teams to land RL sampling infra, value-function, and data changes; reduced timeouts and NaNs, and added reward-hacking mitigations.
- **RL initialization:** Trained SFT checkpoints for RL climbability, using pass@k and entropy metrics to select initializations with the most headroom under RL.
- **Behavior-quality filtering:** Built a trace-filtering pipeline (heuristic, LLM-judge, agentic filters) removing reasoning loops, bash overuse, and inefficient tool use from distillation data, resulting in better-behaved RL inits.
- **SWE agents:** Improved SWE performance in shipped Gemini models via large-scale SFT distillation (SWE-Bench Pro, DeepSWE); built the eval harness and post-trained the model for DeepMind's first SWE-bench push, launched as Jules.
- **Agent data:** Created synthetic RL tasks and SFT trajectories for agent behaviors such as code navigation; coordinated with external vendors to collect process-supervised agent SFT traces.
- CodeGemma: Built executable evaluations for code fill-in-the-middle (FIM) completion.

Aug. 2022 – May 2022 | **Google DeepMind**, Mountain View, CA
Research Engineer Intern | Manager: Chenjie Gu, Mentors: Peter Choy, Miaosen Wang

- Trained AlphaCode-style models for code generation and editing (PR review comment → code edit) for in-IDE developer workflows.
- Implemented and evaluated RL objectives for sample selection and re-ranking in multi-task training.

Jul. 2023 – Sep. 2021 | **Carnegie Mellon University**, Pittsburgh, PA
Research Assistant | Advisor: Prof. Louis-Philippe Morency, Mentor: Chaitanya Ahuja

- Developed continual-learning techniques to prevent catastrophic forgetting in multimodal gesture generation with GANs. [**ICCV'23**]

Aug. 2021 – Sep. 2020 | **Google Research**, Bangalore, India
Research Engineer | Manager: Dr. Partha Talukdar

- Built entity-aware multilingual translation for technical-video subtitles; rated 9/10 by human raters, used to translate 100 hours of NPTEL lectures.
- Trained neural models for temporal parsing of multilingual queries and built an evaluation benchmark.

Aug. 2020 – Aug. 2019 | **Microsoft Research**, Bangalore, India
Research Intern | Managers: Dr. Kalika Bali, Dr. Monojit Choudhury

- Analyzed resource disparity across 2,500 languages and the research trends driving it. [**ACL'20**, 1500+ citations]
- Probed natural language inference capabilities of pre-trained language models (BERT, RoBERTa). [**CoNLL'20**]

Dec. 2018 – Jul. 2018 | **Microsoft Research**, Bangalore, India
Research Intern | Managers: Dr. Navin Goyal, Dr. Monojit Choudhury

- Built neural semantic parsers converting natural language to code (Regex, SQL), competitive with the state of the art.

Education

May 2023 | **Carnegie Mellon University - School of Computer Science**, Pittsburgh, PA, USA
Aug. 2021 | Master of Science in Intelligent Information Systems, Language Technologies Institute | CGPA: 4.12/4.0

Jun. 2019 | **Birla Institute of Technology and Science Pilani**, Goa, India
Aug. 2015 | Bachelor of Engineering (Honors) in Computer Science | CGPA: 8.44/10.00

Skills

Languages Python, C/C++, Shell, SQL
Tools JAX, PyTorch, TensorFlow, NumPy, Pandas, Google Cloud

Selected Publications

- Gemini Team Google: ..., **Pratik Joshi**, ..., "Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities", 2025 [pdf]
- Gemini Team Google: ..., **Pratik Joshi**, ..., "Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context", 2024 [pdf]
- Gemini Team Google: ..., **Pratik Joshi**, ..., "Gemini: A Family of Highly Capable Multimodal Models", 2023 [pdf]. Coverage: Google Blog, NY Times, Wired, TechCrunch
- CodeGemma Team: ..., **Pratik Joshi**, ..., "CodeGemma: Open Code Models Based on Gemma", 2024 [pdf]
- **Pratik Joshi***, Sebastin Santy*, Amar Budhiraja*, Kalika Bali, Monojit Choudhury, "The State and Fate of Linguistic Diversity and Inclusion in the NLP World", **ACL'20** [pdf]. 1500+ citations. Coverage: US FTC Commissioner remarks, Quartz, Ruder
- Chaitanya Ahuja, **Pratik Joshi**, Ryo Ishii, Louis-Philippe Morency, "Continual Learning for Personalized Co-speech Gesture Generation", **ICCV'23** [pdf]
- **Pratik Joshi***, Somak Aditya*, Aalok Sathe*, Monojit Choudhury, "TaxiNLI: Taking a Ride up the NLU Hill", **CoNLL'20** [pdf][code]
- (alphabetical) Basil Abraham, ..., **Pratik Joshi**, ..., Vivek Seshadri, "Crowdsourcing Speech Data for Low-Resource Languages from Low-Income Workers", **LREC'20** [pdf]. Coverage: Indian Express, Microsoft Stories

Selected Talks and Panels

2024 **Lee Language Lab Talk Series** on "The State and Fate of Linguistic Diversity and Inclusion in the NLP World" [video]
2023 **Centre for Democracy and Technology Panel** on "The Nuts and Bolts (and Nuances) of Foundation Models" [video]
2020 **NLP with Friends** on "State and Fate of Linguistic Diversity in the NLP World" [video]

Honors and Awards

2025 **BITS Global 30 under 30** award for Research Leadership [link]

Service and Teaching

Reviewer LREC'26, COLM'24, ICLR'24, LREC-COLING'24, ACL'24, NAACL'24
Teaching Assistant Machine Learning (BITS F464), BITS Pilani Goa: created coding assignments and final project [course material]